

A Machine Learning Approach for Aircraft Engine Fault Prediction

Capstone Project: EMBA Winter batch 24-26

Area of Interest: Decision Sciences

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Problem & Context

Predict engine failures before they happen, so maintenance can be done just in time.

Today's approach

Periodic maintenance is done at fixed time or flight-hour intervals to meet safety and aviation standards

But unplanned maintenance from component failures is costly, and it grounds the aircraft resulting to customer dissatisfaction

Our approach

- Use multi-sensor (IOT) engine data and past failure records to detect failure patterns
- Predict failures early so maintenance can be planned, cost-effective and timely
- Keep aircraft flying more, which improves availability for customers

What we're building: A Machine Learning solution that warns of an upcoming failure early enough to plan maintenance

Why it matters:

Higher aircraft availability for operators

Less unplanned downtime

Lower financial losses

Move from fixed-interval and unplanned maintenance to data-driven, condition-based maintenance

Brief about Aircraft Engine Data

Data is available

- Modern engines have control and monitoring systems that record operational and health parameters throughout flight (**Engine Fault Records**)
- This produces large volumes of telemetry data for tracking performance, spotting anomalies and supporting maintenance decisions
- Data-driven health monitoring now complements traditional maintenance practices

Faults have many sources

- Component degradation
- Sensor failures
- Actuator malfunctions
- Control-system failures

Why early detection matters

- Better operational reliability
- Lower maintenance costs
- Less unplanned aircraft downtime

ENGINE DATA RECORDS

CHIP 1

CLM_PERF

CRU_PERF

CONTROLLER_STATS

CONTROLLER_TIME_AT_TEMP

ENG_RED

ENG_STATS

ENG_PLA_T0

FAULT_RECORDS

IGNI_REC

A_LIFE

B_LIFE

INCI_REC_A

INC_REC_B

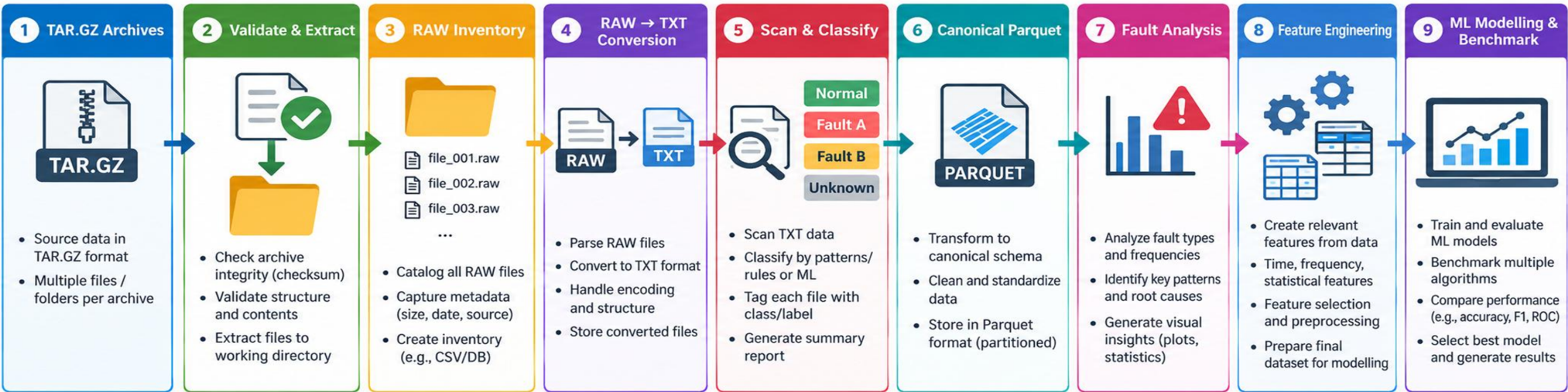
RLDW

STS

TAKEOFF_PERF

Move from fixed-interval and unplanned maintenance to data-driven, condition-based maintenance

End-to-End Pipeline Overview



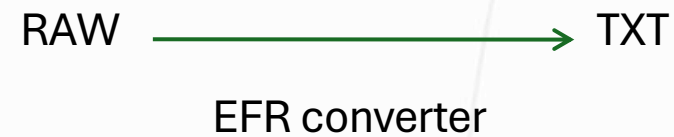
Step Mapping	Task
1 To 3	Data Acquisition and Organization
4 To 6	Data Preprocessing
7	Fault Analysis
8	Feature Engineering
9	ML Algorithms and Model benchmarking

Move from fixed-interval and unplanned maintenance to data-driven, condition-based maintenance

Acquire and Organize Data



- tar.gz file is essentially a archive, The TAR/GZ format is used mainly for packaging and transferring a large number of files efficiently
- After extraction tar.gz file, you get .RAW files, **RAW = original source data**
- The RAW format apparently isn't convenient for directly reading and processing the EFR information.



- The TXT representation exposes the information in a more structured, human/machine-readable form.
- SHA-256 is used for **data integrity and traceability**.

Every Raw/TXT file we have a SHA-256 hash, because out of 1 lakh records, there is a chance that the records get duplicated:

Ex- Engine_001.RAW

↓

SHA-256

↓

A7F82C91.....

That long string is effectively a digital fingerprint or unique ID of the file's contents.

If even a tiny part of the RAW file changes, the hash will normally be completely different.

Identification of faults

`export_fault_metrics_to_excel.py`

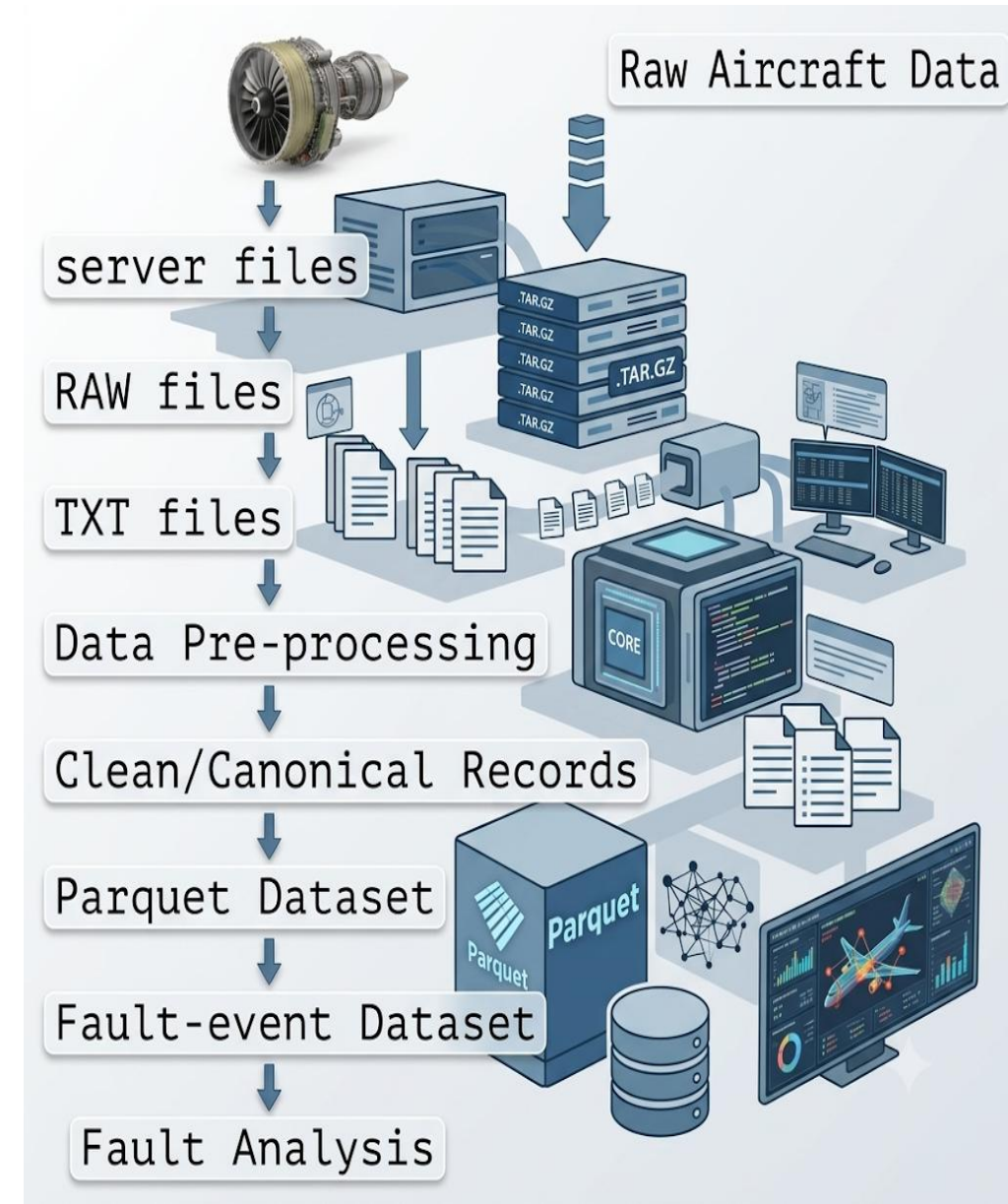
After the aircraft-engine data has already been cleaned and stored in Parquet format.

It reads the processed aircraft-engine fault data from the Parquet file and examines the fault records one by one.

Purpose: Aggregates and analyzes engine fault occurrences.

Key Metrics:

- Groups data by fault code.
- Calculates fault frequency and percentage of total faults.
- Identifies unique affected engines and flight legs.
- Captures first and last observed timestamps.



The program also avoids loading the entire dataset into computer memory at once. It processes the data in batches called folds

Fault Analysis

Rank	Fault Code	What is the fault		Importance	Event Count	Affected Engines
1	111	Local communication failure	Sequence of powerup can cause this fault, doesn't mean actually there is a failure		22866	1614
2	105	Comms failure with air data	Sequence of powerup can cause this fault, doesn't mean actually there is a failure		10547	1167
3	61	Cross Channel comms failure with air data	Sequence of powerup can cause this fault, doesn't mean actually there is a failure		9722	1136
4	110	Bleed air status unavailable	Engine level power on sequence and shutdowns has an effect also indicates potential issues in bleed air flow, not enough physics based evidence		7797	1021
5	175	Insufficient rotation(point at which wheels leave the runway) data	Insufficient data points to make any judgement/ may not actually be a fault		6899	914
6	176	Cross side Insufficient rotation data	Insufficient data points to make any judgement/may not actually be a fault		6779	886
7	206	Failure of Thrust reversal system	The problem is genuine and worth solving, makes economic sense		4490	918
8	108	Loss of HP Bleed-Left	Not a standalone fault; this is only a early warning indicator		3794	255
9	109	Loss of HP Bleed-right	Not a standalone fault; this is only a early warning indicator		3793	254
10	156	ICU solenoid open when TR is not disabled	indicates solenoid fault/ power system failure; could be leading indicator of a TR system failure		1999	514

Most of the top faults are noise

Fault 206 as the only fault flagged as "genuine and worth solving."

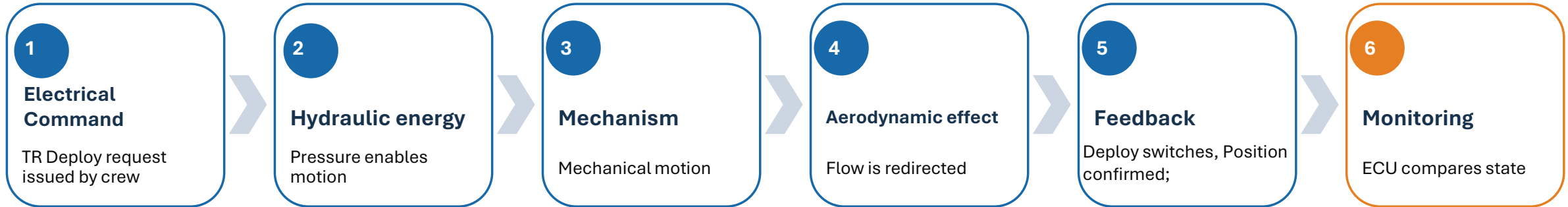
- **Genuine, physical failure mode.** A thrust reverser fault is a real mechanical or hydraulic/electrical issue, so there is a true underlying signal to detect and predict.
- **Operational impact.** The thrust reverser helps decelerate the aircraft on landing. A failure typically leads to maintenance action and hence costs involved.
- **Broad fleet spread.** 918 engines are affected

Fault 156 (ICU solenoid open when TR is not disabled) is a leading indicator of TR failure. Its 1,999 events across 514 engines can serve as a **precursor feature**, giving the model something to predict from.

PHYSICS OF THE PROBLEM

TR functional chain: command to confirmed mechanical state

Deployment is a chain of hydraulic and electrical stages, each with its own status signal and its own timing.



Every stage of this sequence is a measurable timing event.

How long the reverser takes to move from command to locked-and-deployed is, in principle, a direct read on the mechanism's health , the same way a slowing reflex is a symptom before it becomes a failure.

Normal behavior

The commanded state, physical position and feedback indication remain consistent.

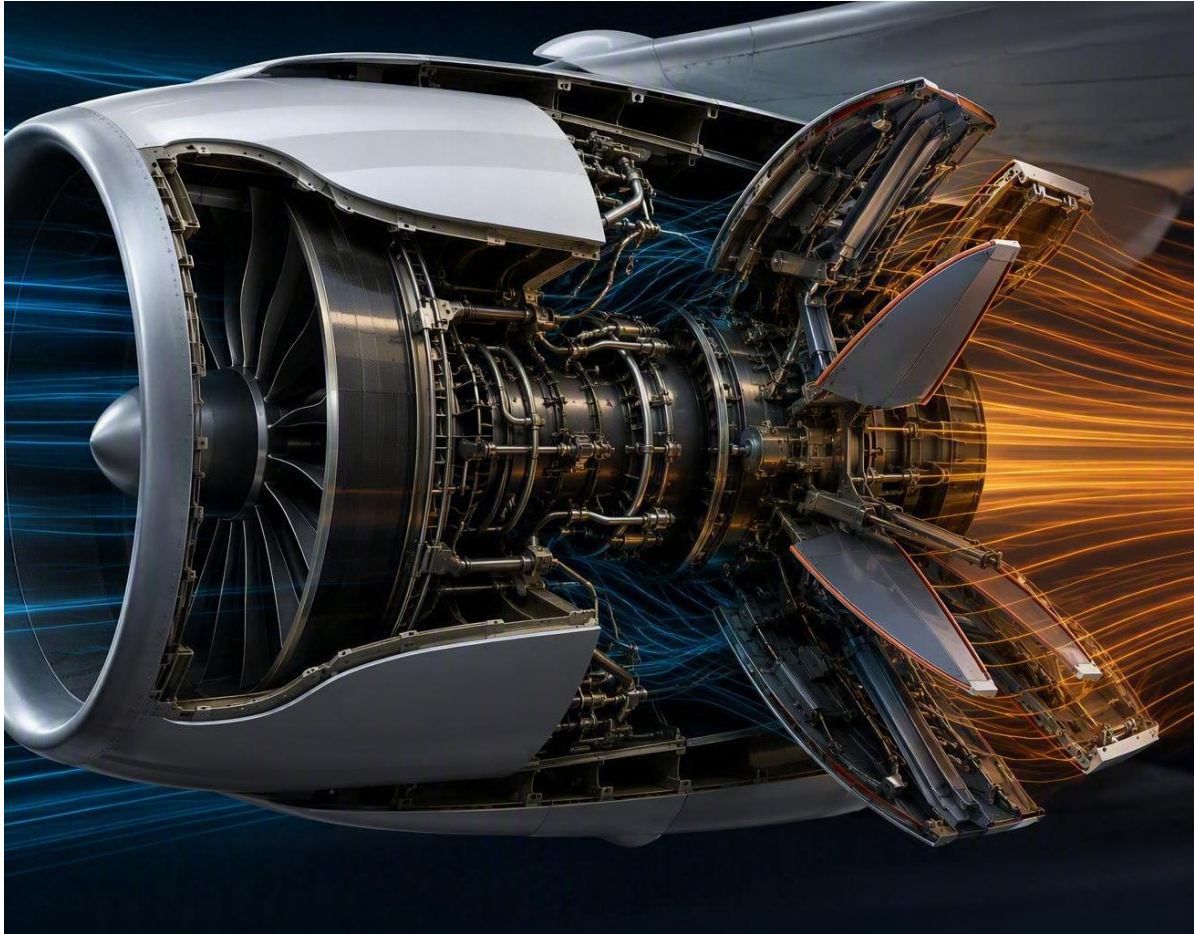
Fault-observable behavior

Command/state disagreement can produce an ENGINE FAULT indication and maintenance conditions in the Engine controller download.

PHYSICS OF THE PROBLEM

Reverse thrust is momentum redirection, not “negative engine thrust”.

The nacelle converts bypass-flow momentum into a decelerating force after touchdown.



Momentum balance

$$F \approx \dot{m} (V_{out} - V_{in})$$

Mechanical wear doesn't appear overnight.

Hydraulic pressure loss, actuator wear, and increased friction accumulate gradually (actuation sequence), these should be measured across a series of landings. It doesn't fail outright or from a clean baseline with no warning.

This is the hypothesis the whole modeling program is built around:

a fault-in-progress should leave a signature in the trend of engine behavior across recent landings, not just in the reading at any single moment.

Data at a glance

98,635

raw Engine Fault Records
partitions processed

1.67M

engine landings in the
landing-level panel

~2,000

engines in the
fleet-wide history

9,300+

candidate features
per landing (V3.2)

The raw data is too large and too wide for convenient repeated CSV processing.

The pipeline therefore converts the information into **Parquet partitions**

Two variants for different models developed are used:

- Top 500 features
- Top 1000 features

The feature-ranking process ranks top 500 or 1000 candidates by means of Pearson correlation.
Absolute Pearson correlation between the feature and the Fault-206 target.

Model development & Feature Engineering

For every 30-day window, the code calculates things such as:

TOTAL_EVENTS_30D

Number of recorded events in the previous 30 days.

The code developed looks for **fault 206 in Failure_RECORDS** with a reliable explicit GMT timestamp.

Then it asks:

“After this snapshot, does fault 206 occur within the next 14 days?”

If yes: FAILS_WITHIN_14D = 1

If no: FAILS_WITHIN_14D = 0

So, the ML problem is modelled as below:

Previous 30 days of engine behaviour

↓

ML model

↓

Will fault 206 occur within next 14 days?

OBS, ACTIVE, LAST, CHANGES

These are **four ways of summarizing a parameter's recent history** over the previous 30 days:

- **OBS** → How many times did we receive a reading?
Example: Oil pressure was recorded 500 times.
- **ACTIVE** → How often was the parameter active/non-zero?
Example: Oil-pressure abnormal condition occurred frequently.
- **LAST** → What was the most recent value?
Example: The latest oil pressure was 50.
- **CHANGES** → How often did the value/status change?
Example: If a status keeps switching ON/OFF, CHANGES will be high.

Comparison of approaches

Approach A

Calendar-Anchored Snapshots

One row per engine, every 7 days, regardless of activity. Each row summarizes the trailing 30 days of behavior.

A 14-day forward-looking horizon (the target this program predicts against) converts a same-day emergency into a schedulable maintenance event

The 14-Day Window

14 days of scheduling flexibility



This is exactly why the program targets "fault within the next 14 days," not "fault right now": a same-day prediction has no economic value — there's no time left to act on it.

Approach B

Landing-Anchored Windows

One row per physical landing. Features drawn from a short window right around that specific landing as TR only acts at landing.

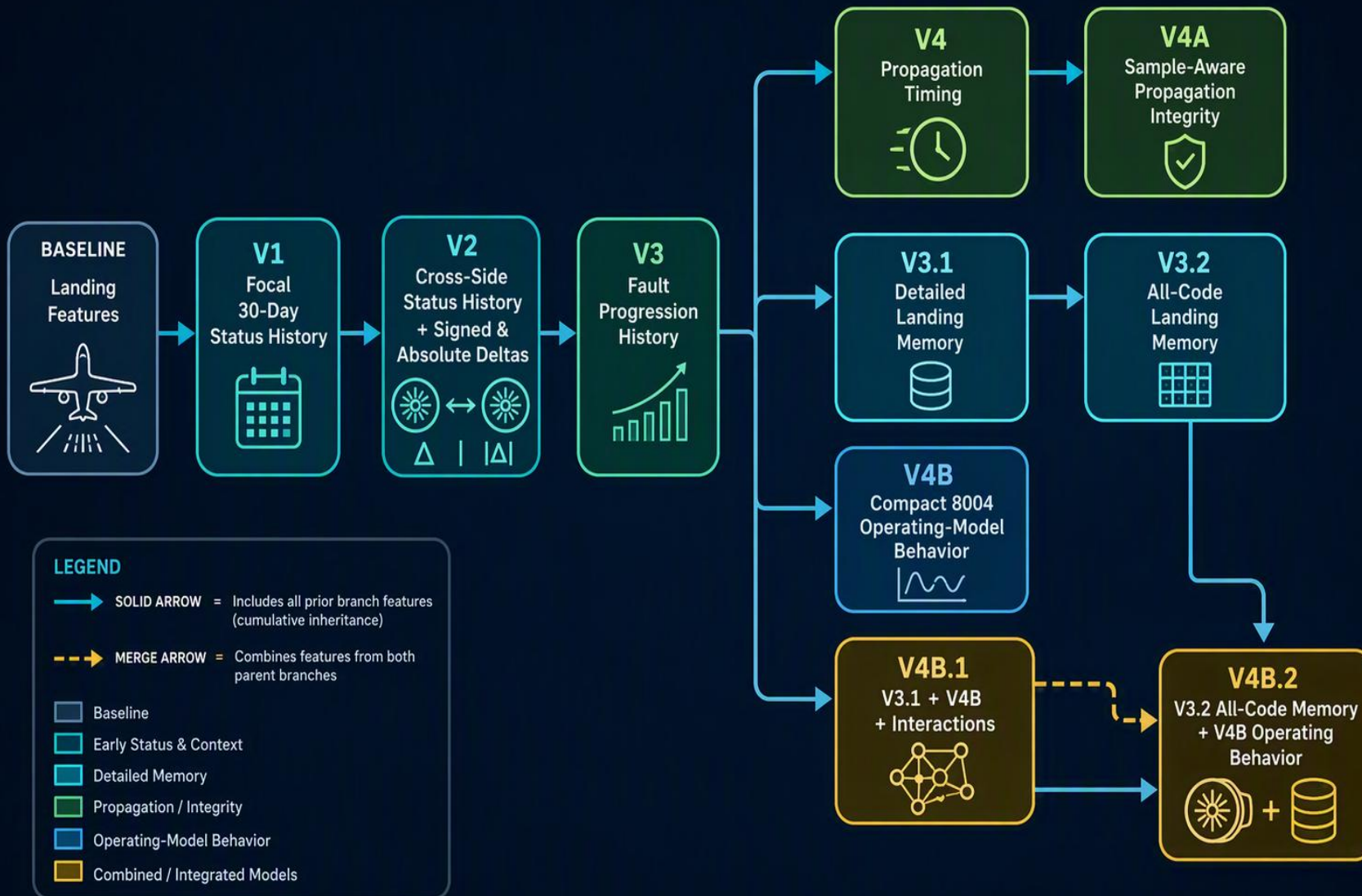
- **One row = one engine landing**
- Features are extracted from a **short window around each landing**
- The model uses the **previous 30 days of engine behaviour** to predict whether a **fault will occur within the next 14 days**
- **Engine-grouped validation** reduces the risk of the model memorizing individual engines

The 7-day frequency is for creating prediction snapshots

Project Roadmap

EVOLUTION OF AIRCRAFT PREDICTIVE-MAINTENANCE FEATURE MODEL

CUMULATIVE & BRANCHING DEVELOPMENT PATHWAYS



Each step builds on all prior information unless merged with a parallel branch.

Version	Incremental code/feature change	What it adds
Baseline	Landing-level input features	Basic landing measurements and derived descriptors
V1	Added 30-day focal-engine history	Previous status/fault behaviour of the same engine
V2	Added cross-side engine information	Other engine on same aircraft + signed/absolute differences
V3	Added fault-progression history	Recent fault occurrence/progression over time
V4	Added propagation timing	Looks at command/response propagation behaviour

ML algorithms & End-to-End architecture

Latest model compares **six model families**:

1. Histogram Gradient Boosting
2. Extra Trees
3. Random Forest
4. Elastic-Net Logistic Regression
5. SGD Logistic Regression
6. XGBoost

So the experiment is now:

28 feature variants (V1,V2,V3,V4 etc., with 500 and 1000 features

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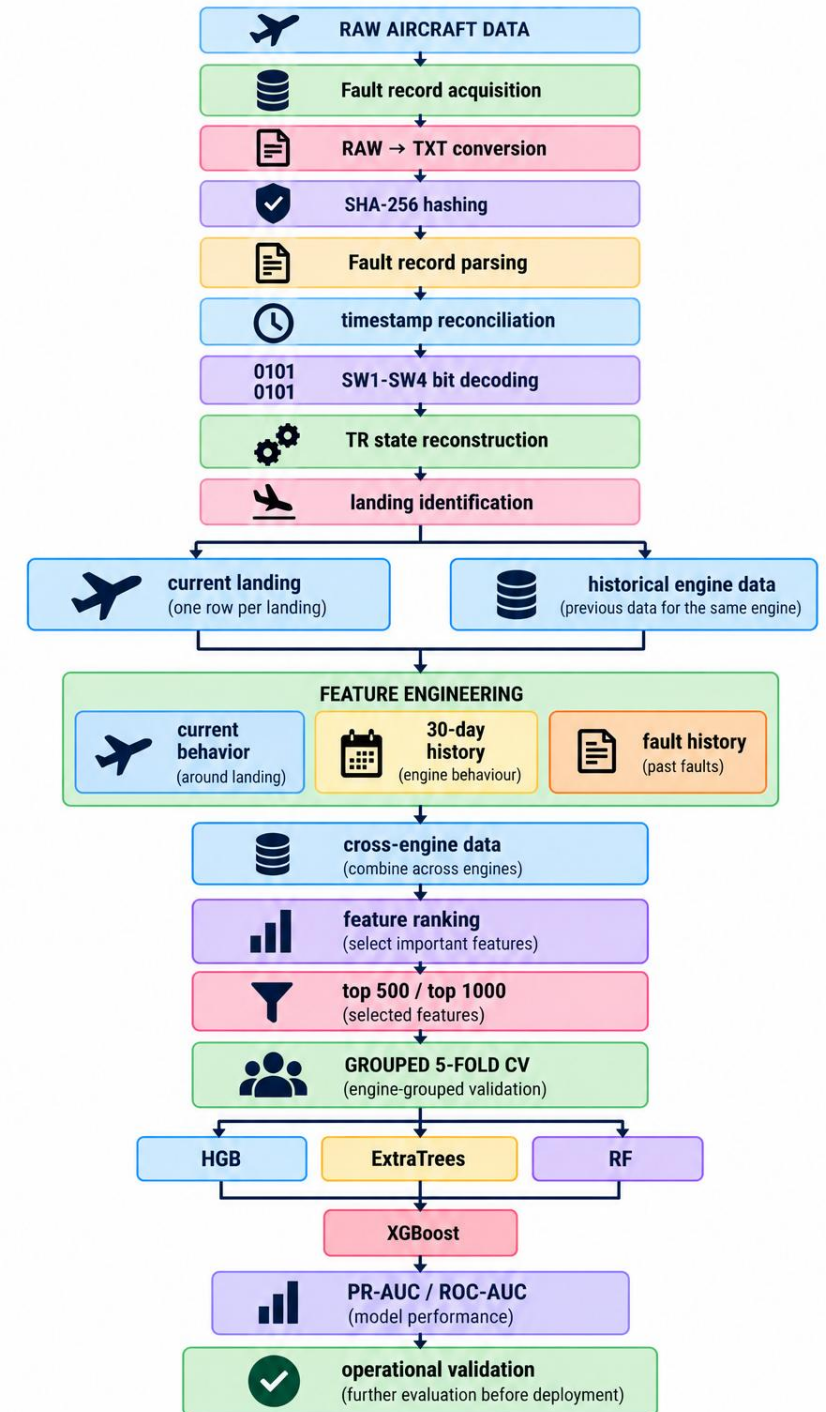
6 ML algorithms

=

168 model experiments

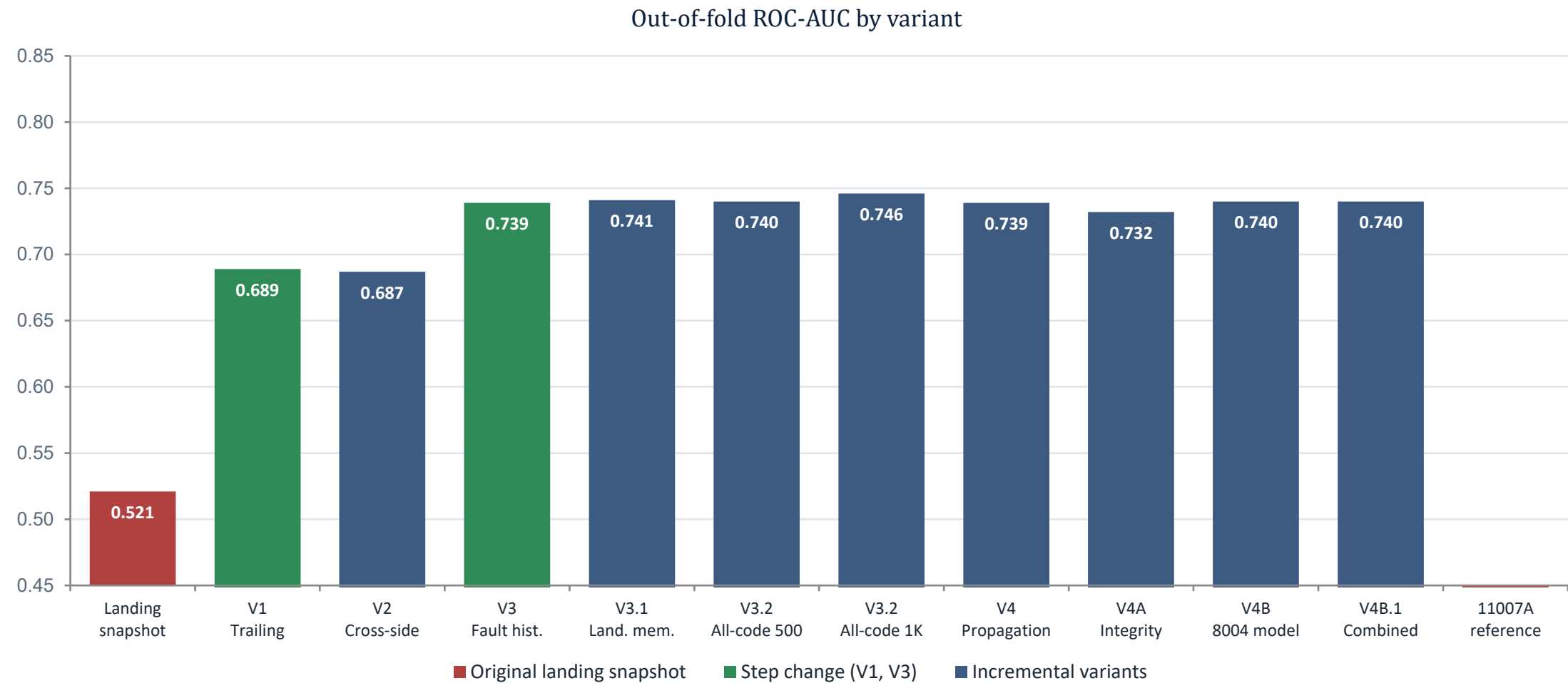
Each experiment uses: **5 validation folds**

So there are effectively: $168 \times 5 = 840$ fold-level evaluations.



Variants with Incremental Progress

Each variant adds one feature family to its parent. Rows (1.67M landings), labels, aircraft-grouped folds, and learner stay matched.



Results

Strongest result we have obtained is

hybrid_v3_2_all_fault_landing_memory_ver05
hybrid_v3_2 feature ranking

Top 1000 with: **Histogram Gradient Boosting**

Another high-performing combination is:

hybrid_v4b_2_8004_all_fault_memory_ver04

Top 1000: **XGBoost**

with: **ROC-AUC $\approx$ 0.734**

and: **PR-AUC $\approx$ 0.123**

Landing-level data + historical fault/memory features + landing features + cross-side engine comparisons $\rightarrow$ feature ranking $\rightarrow$ 500/1000 selected features $\rightarrow$ six ML algorithms $\rightarrow$ grouped aircraft-level validation $\rightarrow$ predict Fault 206 within 14 days.

The strongest reported configuration is currently around **ROC-AUC 0.745** and **PR-AUC 0.123**, with **1.67 million prediction records** and **5.25% positive prevalence**.

Metric	Result
ROC-AUC	0.7446
PR-AUC	0.1229
Prevalence	5.25%
PR Lift	2.34×
Features	995
Records	1,669,294

The expected output is a plain-language, actionable alert — exactly the form this needs to take to be useful to a maintenance planner:

“Tail N12345 — RIGHT engine shows a rising thrust reverser precursor signature over the last 5 landings. Recommend inspection within 14 days.”

Next Steps & Path Forward

Moving from Strategy to Action — Together



Availability

Convert surprise removals into planned maintenance windows



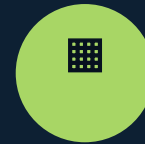
Labor

Reduce manual search and shorten fault isolation



Material

Replace the right component instead of broad troubleshooting



Inventory

Stage spares against measured risk, not blanket assumptions



Customer impact

Reduce delays, cancellations and no-dispatch exposure

Current scenario: cost of unplanned maintenance by fleet scale

Failure-driven costs create material, labour, logistics and operational disruption exposure.

Cost component	Current cost / event	Calculation role
Direct material	\$65,000	Replacement parts after failure
Direct labour	\$20,000	Unplanned maintenance effort
Logistics & freight	\$7,500	Expedited movement of parts
Operational downtime	\$40,000	Unavailable operating time
Passenger & crew accommodation	\$10,000	Disruption-related support
Unplanned repair subtotal	\$142,500	Per failure event

Calculation item	100 engines	4,000 engines
Engines in scope	100	4,000
Expected failures at 5%	5	200
Unplanned repairs	5 × \$142,500 = \$712,500	200 × \$142,500 = \$28,500,000
Periodic inspections / tests	\$180,000	\$7,200,000
Total current annual cost	\$892,500	\$35,700,000

Current Cost Estimates For Unplanned Repair Activities	App Estimated Cost in \$
Periodic Inspections and Test per Engine	20000
Aircraft grounding time and loss of availability cost	25000
	45000

\$712,500

annual unplanned repair cost for 100 engines at 5% failures

5 failures × \$142,500

\$35.7M

CURRENT ANNUAL COST FOR 4,000 ENGINES

200 expected failures annually

Expected savings with predictive planned maintenance



Early intervention reduces direct repair costs and removes the modelled disruption-cost categories.

Cost component	Current / event	Planned / event	Savings / event
Direct material	\$65,000	\$35,000	\$30,000
Direct labour	\$20,000	\$4,000	\$16,000
Logistics & freight	\$7,500	\$2,000	\$5,500
Operational downtime	\$40,000	\$0	\$40,000
Passenger & crew accommodation	\$10,000	\$0	\$10,000
Total	\$142,500	\$41,000	\$101,500

71.2%

cost reduction per repair event

\$101,500 saved per event

Calculation item	4,000 engines
Engines in scope	4,000
Expected failures at 5%	200
Planned repairs	200 × \$41,000 = \$8,200,000
Periodic inspections / tests	\$0
Total annual cost after model deployment to production	\$8,200,000

Current Cost Estimates For Unplanned Repair Activities	App Estimated Cost in \$
Periodic Inspections and Test per Engine	0
Aircraft grounding time and loss of availability cost	0
Total	0

Managerial implications for aircraft engines

Predictive maintenance changes the operating model, not only the cost base.



Maintenance strategy

Shift from failure response to risk-based work planning; schedule interventions within available maintenance windows.

Fleet availability

Reduce modelled downtime and accommodation costs; protect dispatch reliability and asset utilisation.

Parts & workforce

Pre-position material, avoid premium freight and align skilled labour with forecasted engine risk.

Governance control & Regulatory oversight

Track precision, recall, false alerts, avoided failures and realised savings; retain engineering sign-off. Approach to be approved by the regulator(DGCA/FAA)

Management action: validate the assumptions through a controlled pilot.

Future scope of work

Maintenance actions is digitised

Engine logbooks are being digitised and will be fed to the model

Addition of High density data

High Density data will be added to the model and it is expected to get ROC data up 0.8

Integration of borescope inspection Video records

Video records will be used for modelling for predicting the faults

LLM integration on the edge device

Create actionable alerts after every landing

Note: Work underway for the above which have higher impact on engine maintenance philosophy.

Thank you

The background is a solid dark blue. It features several abstract elements: a grid of thin, light blue lines in the upper right corner; a series of concentric, thin, light blue arcs in the lower left corner; and a wavy, dotted line in light blue that spans the bottom of the image. There are also several small, solid blue squares scattered across the background.

The Strongest Natural Experiment Available- CROSS-ENGINE COMPARISON

Two engines on the same aircraft, at the same landing, share almost everything that could otherwise confound a comparison: runway condition, aircraft weight, atmosphere, pilot technique.

That leaves engine health as close to the only variable as real operational data gets — a far cleaner comparison than across different landings or different aircraft.

Suppose an aircraft has:

Engine A = left engine

Engine B = right engine

and both engines are operating on the same aircraft during the same landing.

Work in Progress- to be further explored

